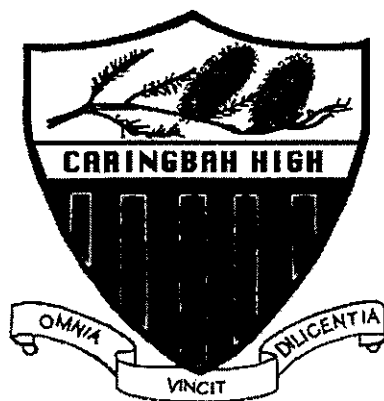




Caringbah High School

Year 12 2019 Mathematics HSC Course Assessment Task 4

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

Section I 10 marks

Attempt Questions 1-10
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 90 marks

Attempt Questions 11-16
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only								
Section I	Section II						Total	
Q 1-10	Q11	Q12	Q13	Q14	Q15	Q16		
/10	/15	/15	/15	/15	/15	/15	/100	%

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1. What is the value of $\log_2 3$, correct to 4 significant figures?

(A) 1.584
(B) 1.585
(C) 0.6309
(D) 0.6310

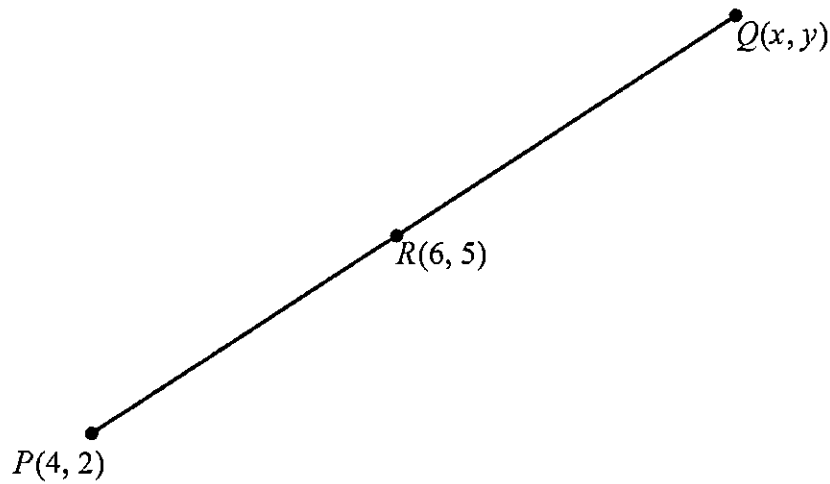
2. The primitive function of $\sec^2 2x$ is...

(A) $\tan x + c$
(B) $\tan 2x + c$
(C) $\frac{1}{2}\tan x + c$
(D) $\frac{1}{2}\tan 2x + c$

3. How many solutions does the equation $(\cos x + 1)(2 \sin x - 1) = 0$ for $0 \leq x \leq 2\pi$ have?

(A) 2
(B) 3
(C) 4
(D) 5

4. The point $R(6, 5)$ is the midpoint of the interval PQ , where P has the coordinates $(4, 2)$. What is the coordinate of Q ?



- (A) $(9, 9)$
(B) $(7, 8)$
(C) $(8, 8)$
(D) $(8, 9)$
5. What are the domain and range of $f(x) = \sqrt{4 - x^2}$?
- (A) Domain: $-2 \leq x \leq 2$ Range: $0 \leq y \leq 2$
(B) Domain: $-2 \leq x \leq 2$ Range: $-2 \leq y \leq 2$
(C) Domain: $0 \leq x \leq 2$ Range: $-4 \leq y \leq 4$
(D) Domain: $0 \leq x \leq 2$ Range: $0 \leq y \leq 4$
6. A particle moves so that its velocity function at time t seconds is given by:

$$v = 2e^{-t}(1 - t)$$

At what time is the acceleration zero?

- (A) $t = 0$
(B) $t = 1$
(C) $t = 2$
(D) $t = 3$

7. Find the perimeter (P) of the sector of a circle with radius 10 cm and an angle of $\frac{\pi}{3}$ subtended at the centre.

(A) $P = 0.5 \times 100 \times \left(\frac{\pi}{3} - \sin \frac{\pi}{3}\right) \text{ cm}$

(B) $P = \left(0.5 \times 100 \times \frac{\pi}{3}\right) \text{ cm}$

(C) $P = \left(20 + \frac{\pi}{3}\right) \text{ cm}$

(D) $P = \left(20 + \frac{10\pi}{3}\right) \text{ cm}$

8. Which of the following is equal to $\frac{1}{3\sqrt{5}+\sqrt{2}}$?

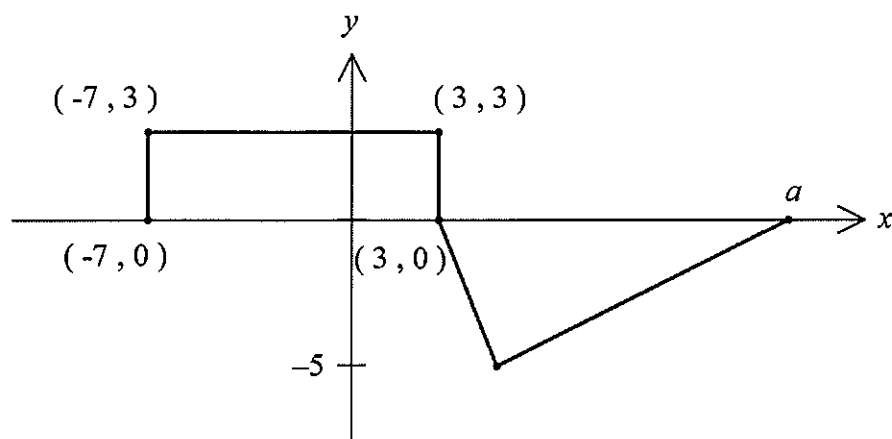
(A) $\frac{3\sqrt{5} - \sqrt{2}}{13}$

(B) $\frac{3\sqrt{5} + 2}{13}$

(C) $\frac{3\sqrt{5} - \sqrt{2}}{43}$

(D) $\frac{3\sqrt{5} + \sqrt{2}}{43}$

9. The diagram shows the graph of $y = f(x)$.



Use the graph to determine the value of a which satisfies the condition

$$\int_{-7}^a f(x) dx = 0.$$

- (A) 9
(B) 12
(C) 13
(D) 15
10. The solution of the equation $\ln(x + 2) - \ln x = \ln 4$ is

- (A) $x = \frac{2}{3}$
(B) $x = \frac{2}{5}$
(C) $x = \frac{3}{2}$
(D) $x = \frac{5}{2}$

End of Section I- Multiple Choice Questions

Section II

60 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Calculate correct to one decimal place, the value of 1

$$\sqrt{\frac{3xy}{z}} \text{ when } x = 4.2, y = 6.8 \text{ and } z = 4.4$$

- (b) Simplify: $\frac{x - x^{-2}}{1 + x^{-2}}$ 2

- (c) Differentiate:

(i) $\frac{2}{5x^3}$ 1

(ii) $e^x \sin x$ 2

(iii) $(e^{3x} - 5)^4$ 2

- (d) A parabola has an equation given by: $y = \frac{1}{2}(x^2 - 6x + 19)$

(i) Express the above equation in the form $(x - h)^2 = 4a(y - k)$. 2

(ii) Find the coordinates of the vertex and focus of the parabola. 2

(iii) Find the equation of the directrix of the parabola. 1

(iv) Sketch the locus of P , indicating all the above features. 2

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the primitive of $\cos(2x + 1)$. 1
- (b) If α and β are the roots of the equation $2x^2 - 3x + 4 = 0$,
find the value of $\alpha^2 + \beta^2$. 3
- (c) Find, in general form, the equation of the tangent to the curve $y = x \ln x$ at the
 x – intercept. 3
- (d) Find $\int_0^1 \frac{3x}{x^2 + 1} dx$, in exact form. 3
- (e) Solve and graph the solution for $|1 - 3x| \geq 2$. 3
- (f) For what values of k will the equation $x^2 - kx + 9 = 0$ have real and different
roots? 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Find

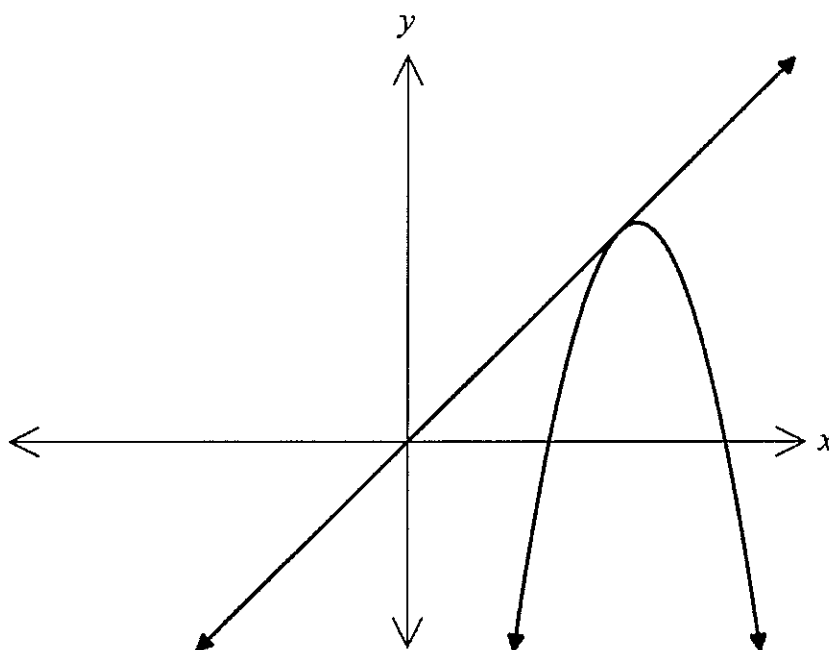
(i) $\int e^{5x+1} dx$ 1

(ii) $\int \cot x dx$ 1

(iii) $\int (x^2 - 1)^2 dx$ 2

(b) Evaluate $\int_0^3 \sqrt{5x+1} dx$. 3

(c) The parabola $y = -x^2 + 13x - 36$, passes through the point $P(6, 6)$.



(i) Show the equation of the tangent to the parabola at the point $P(6, 6)$ is $y = x$. 2

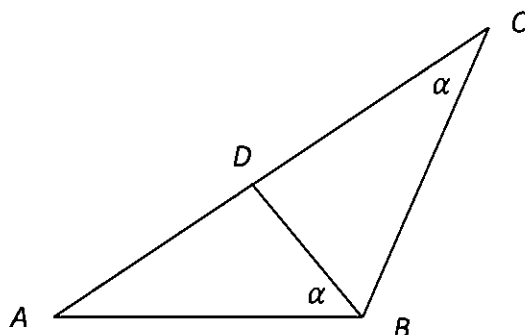
(ii) Find the area bounded by the parabola, the tangent and the x - axis. 4

(d) Solve $2^{2x} - 5 \times 2^x + 4 = 0$ 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a)



(i) Prove that $\triangle ABC$ and $\triangle ADB$ are similar. 2

(ii) If $AD = 4$ cm and $DC = 12$ cm, find the length of AB . 1

(b) Let $f(x) = \frac{1}{3}x^3 + x^2 - 3x + 5$.

(i) Find the stationary points and determine their nature. 4

(ii) Find any inflexion points. 2

(iii) Sketch the graph of $f(x)$. 1

(iv) For what value of x is $f(x)$ concave up? 1

(c) The mass, M , in grams of radioactive substance is expressed as $M = 195e^{-kt}$ where k is a positive constant and t the time in days.
The mass of the substance halved in 6 days.

(i) Find the value of k , correct to 4 decimal places. 2

(ii) At what rate is the mass decaying after 15 days. Answer correct to one decimal place. 2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) Amelia is saving for a holiday. In the first month she saves \$30, in the second month her savings are \$5 more than the month before.
- (i) How much will she save in the 21st month? 1
 - (ii) How much money will she have saved in total by the 21st month? 2
 - (iii) Amelia needs \$2100 to pay for her plane ticket. How long will it take her to save this amount? 2

- (b) Calculate the volume of a solid of revolution if $y = \frac{1}{\sqrt{x}}$ is rotated about the y – axis from $y = 1$ to $y = 2$. 2

- (c) A particle is travelling in a straight line, starting from the origin, such that

$$\frac{d^2x}{dt^2} = \frac{8}{(t+1)^2}$$

where x is displacement in metres and t is time in seconds.

- (i) Explain why the acceleration is always positive. 1
 - (ii) Find the expression for velocity of the particle. 2
 - (iii) Find the distance covered between $t = 2$ and $t = 5$ seconds. 3
- (d) Solve $3 \tan 2x = \sqrt{3}$ for $0 \leq x \leq 2\pi$ 2

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) Sam borrowed \$18 000 to buy a new car. He is charged interest at 12% p.a. compounded monthly, on the balance owing. The loan is to be repaid in equal monthly instalments over 5 years.

Let A_n be the amount owing after the n th monthly repayments M has been made.

- (i) Write an expression for the amount owing after one month, A_1 . 1

- (ii) Show that $A_n = 18000(1.01)^n - 100M(1.01^n - 1)$ 3

- (iii) Calculate Sam's monthly instalment. 1

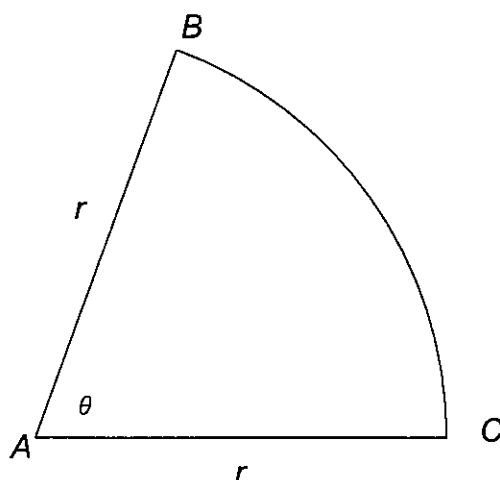
- (b) If $2x^2 - 3x + 3 \equiv A(x - 1)^2 + B(x - 1) + C$, find the value of A, B and C . 2

- (c) Use Simpson's Rule with 3 function values to estimate 2

$$\int_0^{\frac{\pi}{4}} \sec x \, dx$$

Answer correct to two decimal places.

- (d) AB and AC are radii of the length r metres of a circle with centre A . The arc BC of the circle subtends an angle of θ at A . The perimeter of figure ABC is 8 m.



- (i) Show that the area $A \text{ m}^2$ of the sector ABC is given by $A = \frac{32\theta}{(2+\theta)^2}$. 2

- (ii) Hence, find the maximum area of the sector. 4

End of Paper!

Section (I)

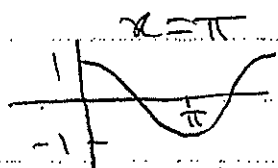
Q1	B	Q6	C
Q2	D	Q7	D
Q3	B	Q8	C
Q4	C	Q9	D
Q5	A	Q10	A

$$\text{Q1. } \log_2 3 = \frac{\log_e 3}{\log_e 2} = 1.585 (B)$$

Ex 2 $\int \sec^2 2x \, dx = \frac{1}{2} \tan 2x + C$

Q3 $(\cos x + 1)(2\sin x - 1) = 0$ $0 \leq x \leq 2\pi$

$$\cos x = -1 \quad 2 \sin x = 1$$



$$\sin x = \frac{1}{2}$$

$x = \pi/b$

$\frac{5\pi}{b}$

S A T C

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \pi$$

∴ 3 solutions (B)

Q4 $\frac{4+x}{2} = 6, \frac{y+2}{2} = 5$

$$\begin{array}{rcl} 4 + x & = & 12 \\ x & = & 8 \end{array} \qquad \begin{array}{rcl} y + 2 & = & 10 \\ y & = & 8 \end{array}$$

OR

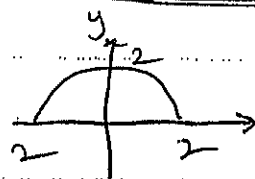
by inspection.

(8, 8) (c)

Q5 $f(x) = \sqrt{4-x^2}$

$$D: -2 \leq x \leq 2$$

$$R: 0 \leq y \leq 2(A)$$

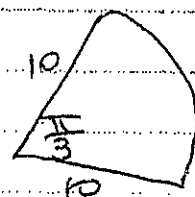


Q6 $v = 2e^{-t}(1-t)$

$$\begin{aligned} a &= 2e^{-t}(-1) + (1-t)(-2e^{-t}) \\ &= -2e^{-t} - 2e^{-t} + 2te^{-t} \\ &= -4e^{-t} + 2te^{-t} \\ &= 2e^{-t}(-2+t) \\ 2e^{-t}(-2+t) &= 0 \\ t &= 2 \quad (c) \end{aligned}$$

Q7 $P = r\theta + 2r$

$$= 10 \times \frac{\pi}{3} + 2 \times 10$$



$$= \left(\frac{10\pi}{3} + 20 \right) \text{ cm (d)}$$

$$\begin{aligned} \frac{Q8}{\frac{1}{3\sqrt{5} + \sqrt{2}} \times \frac{3\sqrt{5} - \sqrt{2}}{3\sqrt{5} - \sqrt{2}}} \\ = \frac{3\sqrt{5} - \sqrt{2}}{45 - 2} \\ = \frac{3\sqrt{5} - \sqrt{2}}{43} \quad (C) \end{aligned}$$

Q9 A rectangle = $3 \times 10 = 30 \text{ u}^2$

$$\begin{aligned} \text{A triangle} &= \frac{1}{2} \times 6 \times 5 \\ &= \frac{56}{2} \end{aligned}$$

$$A_1 = A_2, \text{ for } \int_a^b f(x) dx = 0.$$

$$\frac{56}{2} = 30$$

$$5b = 60$$

$$b = 12$$

$$a = b + 3$$

$$= 12 + 3$$

$$\therefore a = 15 \text{ (D)}$$

Q10 $\ln(x+2) - \ln x = 4$

$$\frac{x+2}{x} = 4$$

$$x+2 = 4x$$

$$3x = 2$$

$$x = \frac{2}{3} \text{ (A)}$$

Section II

Q11

(a) $\sqrt{\frac{3xy}{z}} = \sqrt{\frac{3 \times 4 \cdot 2 \times 6 \cdot 8}{4 \cdot 4}}$

$$= 4.4$$

(b) $\frac{x - x^{-2}}{1 + x^{-2}} = \frac{x - \frac{1}{x^2}}{1 + \frac{1}{x^2}}$

$$= \frac{x^3 - 1}{x^2} \div \frac{x^2 + 1}{x^2}$$

$$= \frac{x^3 - 1}{x^2} \times \frac{x^2}{x^2 + 1}$$

$$= \frac{x^3 - 1}{x^2 + 1}$$

(c) (i) $\frac{d}{dx} \left(\frac{2}{5x^3} \right) = \frac{d}{dx} \left(\frac{2}{5} x^{-3} \right)$

$$= -\frac{6}{5} x^{-4}$$

$$= -\frac{6}{5x^4}$$

(ii) $\frac{d}{dx} (e^x \sin x)$
 $= e^x \cos x + \sin x e^x$
 $= e^x (\cos x + \sin x)$

(iii) $\frac{d}{dx} (e^{3x} - 5)^4$

$$= 4(e^{3x} - 5)^3 (3e^{3x})$$

$$= 12e^{3x} (e^{3x} - 5)^3$$

(d) (i) $y = \frac{1}{2}(x^2 - 6x + 19)$

$$2y = x^2 - 6x + 19$$

$$2y = x^2 - 6x + 9 + 10$$

$$2y - 10 = (x - 3)^2$$

$$2(y - 5) = (x - 3)^2$$

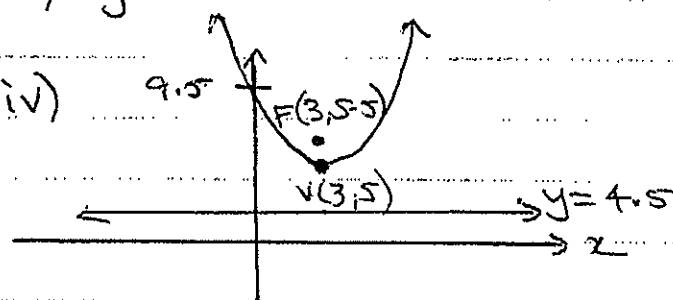
$$\therefore (x - 3)^2 = 2(y - 5)$$

(ii) $a = \frac{1}{2}$

$$\therefore V(3, 5) \quad F(3, 5.5)$$

(iii) $y = 4.5$

(iv)



Q12

(a) $\int \cos(2x+1) dx$

$$= \frac{1}{2} \sin(2x+1) + C$$

(b) $2x^2 - 3x + 4 = 0$

$$a+b = \frac{3}{2}, \quad ab = 2$$

$$a^2 + b^2 = (a+b)^2 - 2ab$$

$$= \left(\frac{3}{2}\right)^2 - 2(2)$$

$$= -\frac{5}{4}$$

Q12 cont.

$$\begin{aligned} \text{(c)} \quad y &= x \ln x \\ y' &= x\left(\frac{1}{x}\right) + \ln x(1) \\ &= 1 + \ln x. \end{aligned}$$

$$x \rightarrow \text{int}, y \rightarrow \infty.$$

$$x \ln x = 0.$$

$$x = 0, 1$$

The only tangent that can exist is at $(1, 0)$

$$\text{when } x=1, y' = 1 + \ln(1)$$

$$\therefore m = 1$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 1)$$

$$y = x - 1$$

$$x - y - 1 = 0.$$

$$\text{(d)} \int_0^1 \frac{3x}{x^2+1} dx$$

$$= \frac{3}{2} \int_0^1 \frac{2x}{x^2+1} dx.$$

$$= \frac{3}{2} [\ln(x^2+1)]_0^1$$

$$= \frac{3}{2} [\ln 2 - \ln 1]$$

$$= \frac{3}{2} [\ln 2 - 0]$$

$$= \frac{3}{2} \ln 2$$

$$\text{(e)} |1-3x| \geq 2$$

$$1-3x = 2$$

$$-3x = 1$$

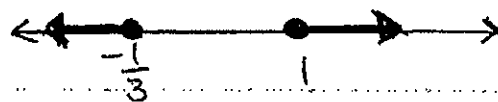
$$x = -\frac{1}{3}$$

$$1-3x = -2$$

$$-3x = -3$$

$$x = 1$$

test



$$\therefore x \leq -\frac{1}{3}, x \geq 1$$

$$\text{(f)} x^2 - kx + 9 = 0$$

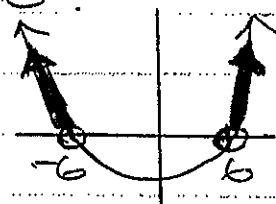
For real & different roots:

$$\Delta > 0$$

$$(-k)^2 - 4(1)(9) > 0$$

$$k^2 - 36 > 0$$

$$k^2 > 36$$



$$\therefore k < -6, k > 6$$

Q13

$$\text{(a) (i)} \int e^{5x+1} dx$$

$$= \frac{1}{5} e^{5x+1} + C$$

$$\text{(ii)} \int \cot x dx$$

$$= \int \frac{\cos x}{\sin x} dx$$

$$= \ln |\sin x| + C$$

$$\text{(iii)} \int (x^2-1)^2 dx$$

$$= \int (x^4 - 2x^2 + 1) dx$$

$$= \frac{x^5}{5} - \frac{2x^3}{3} + x + C$$

Q13 cont

$$(b) \int_0^3 \sqrt{5x+1} dx$$

$$= \int_0^3 (5x+1)^{1/2} dx$$

$$= \left[\frac{2(5x+1)^{3/2}}{3 \times 5} \right]_0^3$$

$$= \frac{2}{15} \left[(5(3)+1)^{3/2} - 1 \right]$$

$$= \frac{2}{15} \left[16^{3/2} - 1 \right]$$

$$= \frac{2}{15} \left[64 - 1 \right]$$

$$= \frac{2}{15} \times 63$$

$$= \frac{42}{15}$$

$$= 8 \frac{2}{5}$$

$$(c) (i) y = -x^2 + 13x - 36 \quad p(6, 6)$$

$$y' = -2x + 13$$

$$\text{sub } x=6, y' = -2(6) + 13$$

$$= -12 + 13$$

$$\therefore m = 1$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = 1(x - 6)$$

$$y = x$$

$$(ii) \text{ For } y = -x^2 + 13x - 36$$

$$x\text{-int, } y=0$$

$$x^2 - 13x + 36 = 0$$

$$(x-9)(x-4) = 0$$

$$x = 9, 4$$

$$A = \frac{1}{2}bh - \int_4^9 \text{parabola}$$

$$= \frac{1}{2} \times 6 \times 6 - \int_4^9 (-x^2 + 13x - 36) dx$$

$$= 18 - \left[\frac{-x^3}{3} + \frac{13x^2}{2} - 36x \right]_4^9$$

$$= 18 - \left[\left(\frac{-6^3}{3} + \frac{13(6)^2}{2} - 36(6) \right) - \left(\frac{-4^3}{3} + \frac{13(4)^2}{2} - 36(4) \right) \right]$$

$$= 18 - \left[-54 - \left(-\frac{184}{3} \right) \right]$$

$$= 18 + 54 - \frac{184}{3}$$

$$= \frac{32}{3}$$

$$= 10 \frac{2}{3} u^2$$

$$(d) 2^{2x} - 5 \times 2^x + 4 = 0$$

$$(2^x)^2 - 5 \times 2^x + 4 = 0$$

$$\text{let } u = 2^x$$

$$u^2 - 5u + 4 = 0$$

$$(u-4)(u-1) = 0$$

$$u = 4, 1$$

$$\text{But } u = 2^x$$

$$2^x = 4$$

$$2^x = 1$$

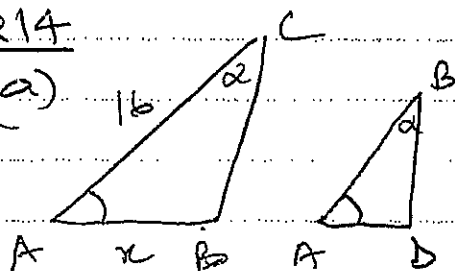
$$2^x = 2^2$$

$$2^x = 2^0$$

$$\therefore x = 2, 0$$

Q14

(a)



(i) In $\triangle ABC$ & $\triangle ADB$,
 $\angle ACB = \angle ADB = \alpha$ (given)
 $\angle CAB = \angle BAD$ (common $\angle$)
 $\therefore \triangle ABC \parallel \triangle ADB$ (equiangular)

(ii) $\frac{AB}{AD} = \frac{AC}{AB}$

$$\frac{x}{4} = \frac{16}{x}$$

$$x^2 = 64$$

$$x = 8 \text{ (as } x > 0)$$

b) (i) $f(x) = \frac{1}{3}x^3 + x^2 - 3x + 5$

$$f'(x) = x^2 + 2x - 3$$

$$f'(x) = 0, \text{ stat. pt.}$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3, 1$$

$$f(-3) = 14, f(1) = \frac{10}{3}$$

$$f''(x) = 2x + 2$$

$$f''(-3) = 2(-3) + 2$$

$$= -4 < 0, \text{ max}$$

$$f''(1) = 2(1) + 2$$

$$= 4 > 0 \text{ min}$$

$\therefore (-3, 14)$ is a max pt. &

$(1, \frac{10}{3})$ is a min pt.

(ii) $f''(x) = 2x + 2$

$$f''(x) = 0, \text{ inflexion pt.}$$

$$2x + 2 = 0$$

$$2x = -2$$

$$x = -1$$

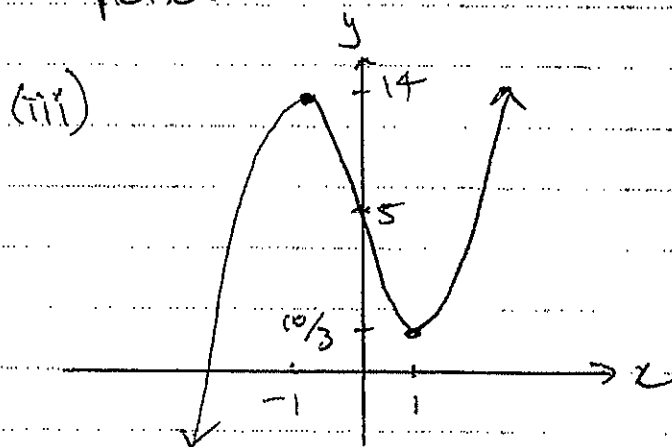
$$f(-1) = \frac{26}{3}$$

$\therefore (-1, \frac{26}{3})$ is a possible inflexion point.

x	-2	-1	0
$f''(x)$	-2	0	2

$\therefore$ change in concavity

$\therefore (-1, \frac{26}{3})$ is an inflexion point.



(iv) $x > -1$

(c) (i) $M = 195e^{-kt}$

when $t=0, M=195$

$$97.5 = 195e^{-6k}$$

$$0.5 = e^{-6k}$$

$$\ln(0.5) = \ln e^{-6k}$$

$$-6k = \ln(0.5)$$

$$k = \frac{-\ln(0.5)}{6} = 0.1155$$

Q14 cont.

$$\begin{aligned} \text{(c) (ii) } \frac{dM}{dt} &= -195ke^{-kt} \\ &= -195(0.1155)e^{-0.1155(15)} \\ &= -3.98... \\ &\approx -4.0 \text{ g/day} \end{aligned}$$

Q15

(a) (i) 30, 35, 40, ...

$$\begin{aligned} T_{21} &= 30 + 20 \times 5 \\ &= \$130 \end{aligned}$$

$$\begin{aligned} \text{(ii) } S_{21} &= \frac{21}{2} (30 + 130) \\ &= \$1680 \end{aligned}$$

$$\text{(iii) } 2100 = \frac{n}{2} [2(30) + (n-1)5]$$

$$4200 = n [60 + 5n - 5]$$

$$4200 = n [5n + 55]$$

$$4200 = 5n^2 + 55n$$

$$5n^2 + 55n - 4200 = 0$$

$$5(n^2 + 11n - 840) = 0$$

$$n^2 + 11n - 840 = 0$$

$$(n+35)(n-24) = 0$$

$$n = -35, 24$$

$$\therefore n = 24 \text{ since } n > 0$$

$$\therefore 24 \text{ months to pay } \$2100$$

$$\text{(b) } y = \frac{1}{\sqrt{x}}, y=1, y=2$$

$$\begin{aligned} V &= \pi \int_a^b x^2 dy \\ &= \pi \int_1^2 y^{-4} dy \\ &= \pi \left[\frac{y^{-3}}{-3} \right]_1^2 \\ &= -\frac{\pi}{3} \left[\frac{1}{y^3} \right]_1^2 \\ &= -\frac{\pi}{3} \left[\frac{1}{2^3} - \frac{1}{1^3} \right] \\ &= -\frac{\pi}{3} \left[\frac{1}{8} - 1 \right] \\ &= -\frac{\pi}{3} \left[-\frac{7}{8} \right] \\ &= \frac{7\pi}{24} u^3 \end{aligned}$$

$$\begin{aligned} y^2 &= \frac{1}{x} \\ x &= \frac{1}{y^2} \\ x^2 &= \frac{1}{y^4} \\ x^2 &= y^{-4} \end{aligned}$$

$$\text{(c) (i) } \frac{d^2x}{dt^2} = \frac{8}{(t+1)^2}$$

Since both numerator & denominator are both positive, then $\frac{d^2x}{dt^2} > 0$.

Q15 cont.

$$(c)(ii) \frac{dx}{dt} = 8 \int (t+1)^{-2} dt$$

$$= 8 \left[\frac{(t+1)^{-1}}{-1(1)} \right] + C$$

$$\frac{dx}{dt} = \frac{-8}{t+1} + C$$

when $t=0$, $\frac{dx}{dt} = 0$.

$$\frac{-8}{0+1} + C = 0$$

$$0+1$$

$$-8 + C = 0$$

$$C = 8$$

$$\therefore \frac{dx}{dt} = \frac{-8}{t+1} + 8$$

$$(iii) x = \int_2^5 \left(\frac{-8}{t+1} + 8 \right) dt$$

$$= -8 \int_2^5 \frac{1}{t+1} dt + \int_2^5 8 dt$$

$$= -8 \left[\ln(t+1) \right]_2^5 + \left[8t \right]_2^5$$

$$= -8 \left[\ln 6 - \ln 3 \right] + \left[8(5) - 8(2) \right]$$

$$= -8 \left[\ln 6 - \ln 3 \right] + (40 - 16)$$

$$= -8 \ln 2 + 24$$

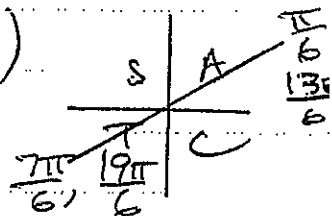
$$= 24 - 8 \ln 2$$

$$(d) 3 \tan 2x = \sqrt{3} \quad 0 \leq x \leq 2\pi$$

$$\tan 2x = \frac{\sqrt{3}}{3} \quad 0 \leq 2x \leq 4\pi$$

$$2x = \tan^{-1} \left(\frac{\sqrt{3}}{3} \right)$$

$$= \frac{\pi}{6}$$



$$\therefore 2x = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}, \frac{19\pi}{6}$$

$$x = \frac{\pi}{12}, \frac{7\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12}$$

Q16

(a) \$18000, 12\% p.a., 5 yrs
= 1% / month, 60 months

$$(i) A_1 = 18000(1.01) - M$$

$$(ii) A_2 = A_1(1.01) - M$$

$$= [18000(1.01) - M](1.01) - M$$

$$= 18000(1.01)^2 - M(1.01) - M$$

$$= 18000(1.01)^2 - M(1.01 + 1)$$

$$A_3 = A_2(1.01) - M$$

$$= [18000(1.01)^2 - M(1.01 + 1)](1.01) - M$$

$$= 18000(1.01)^3 - M(1.01^2 + 1.01) - M$$

$$= 18000(1.01)^3 - M(1 + 1.01 + 1.01^2)$$

$$A_n = 18000(1.01)^n - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{n-1})$$

$$= 18000(1.01)^n - M \left[\frac{1(1.01^n - 1)}{0.01} \right]$$

$$= 18000(1.01)^n - 100M(1.01^n - 1)$$

Q16 cont.

$$(a)(iii) A_{60} = 18000(1.01)^{60} - 100M(1.01^{60} - 1)$$

$$\text{But } A_{60} = 0$$

$$18000(1.01)^{60} - 100M(1.01^{60} - 1) = 0$$

$$100M(1.01^{60} - 1) = 18000(1.01)^{60}$$

$$M = \frac{18000(1.01)^{60}}{100(1.01^{60} - 1)}$$

$$\therefore M = \$400.40$$

$$(b) 2x^2 - 3x + 3 \equiv A(x-1)^2 + B(x-1) + C$$

RHS:

$$A(x-1)^2 + B(x-1) + C$$

$$= A(x^2 - 2x + 1) + Bx - B + C$$

$$= Ax^2 - 2Ax + A + Bx - B + C$$

$$= Ax^2 + (-2A + B)x + A - B + C$$

Equating LHS & RHS:

$$A = 2, -2A + B = -3, A - B + C = 3$$

$$-2(2) + B = -3 \quad 2 - 1 + C = 3$$

$$-4 + B = -3 \quad 1 + C = 3$$

$$B = 1$$

$$C = 2$$

$$\therefore A = 2, B = 1, C = 2$$

$$(c) \int_0^{\pi/4} \sec x \, dx$$

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$
$\sec x$	1	$\sec \frac{\pi}{8}$	$\sqrt{2}$

$$A \doteq \frac{\pi/8}{3} \left[1 + 4 \times \sec \frac{\pi}{8} + \sqrt{2} \right]$$

$$= 0.88$$

$$(d)(i) P = 2r + r\theta$$

$$8 = 2r + r\theta$$

$$8 = r(2 + \theta)$$

$$r = \frac{8}{2 + \theta} \quad \text{--- (1)}$$

$$A = \frac{1}{2} r^2 \theta \quad \text{--- (2)}$$

sub (1) into (2)

$$A = \frac{1}{2} \left(\frac{8}{2 + \theta} \right)^2 \times \theta$$

$$= \frac{1}{2} \times \frac{64}{(2 + \theta)^2} \times \theta$$

$$= \frac{32\theta}{(2 + \theta)^2}$$

$$(ii) \frac{dA}{d\theta} = \frac{(2 + \theta)^2 \times 32 - 32\theta[2(2 + \theta)(1)]}{(2 + \theta)^4}$$

$$= \frac{32(2 + \theta)^2 - 64\theta(2 + \theta)}{(2 + \theta)^4}$$

$$= \frac{32(2 + \theta)[2 + \theta - 2\theta]}{(2 + \theta)^4}$$

$$= \frac{32[2 - \theta]}{(2 + \theta)^3}$$

For max area, $\frac{dA}{d\theta} = 0$.

$$\frac{32(2 - \theta)}{(2 + \theta)^3} = 0, \theta \neq -2$$

$$32(2 - \theta) = 0$$

$$\theta = 2$$

To check nature:

θ	0	2	3
$\frac{dA}{d\theta}$	8	0	$-\frac{32}{125}$

$\therefore \theta = 2$ is a max.

$$\therefore \text{Max. area} = \frac{32(2)}{(2 + 2)^2} = 4 \text{ m}^2$$